

DPR Lecture: Neurologic Exam 1: Mental Status, Cranial Nerves and the Eye Examination

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Make.
Heal.

Innovate.

Reinvent the Future.

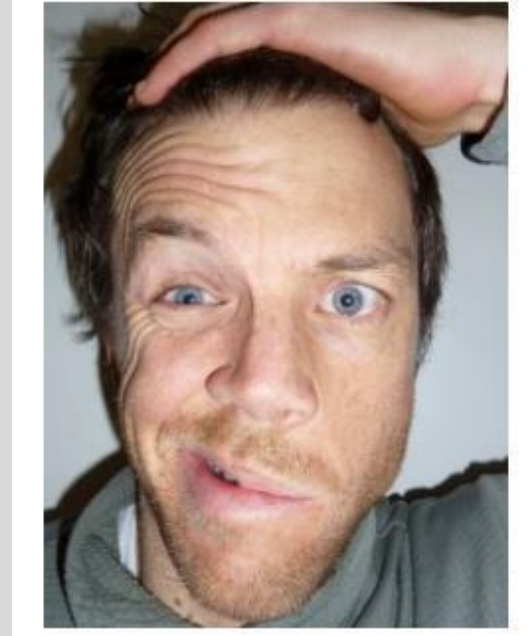
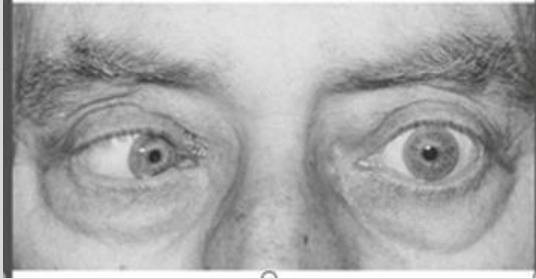
Session Objectives:

- Learn how to obtain a health history in a patient with a neurological complaint
- Assess Mental Status
- Review the Cranial Nerves
- Learn how to perform a Cranial Nerve examination
- Review the eye examination
- Correlate a neurological diagnosis from a Cranial Nerve examination
- Document physical findings for a Cranial Nerve examination and Eye examination

Why Perform a Neurological Exam?

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Components of a Screening Neurological Examination

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- **Mental Status**—level of alertness, appropriateness of responses, orientation to date and place
- **Cranial Nerves**
 - Visual Acuity
 - Pupillary light reflex
 - Eye movements
 - Hearing
 - Facial strength- smile, eye closure
 - Speech
- **Motor System**
 - Strength—shoulder abduction, elbow flexion/extension, wrist extension, finger abduction, hip flexion, knee flexion/extension, ankle dorsiflexion
- **Gait**—casual and tandem
- **Coordination**—fine finger movements, finger-to-nose or finger to chin
- **Reflexes**- Deep Tendon Reflexes and Plantar responses
- **Sensory System**—one modality at toes—can be light touch, pain/temperature, or proprioception

Neurologic Health History

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Common or Concerning Symptoms

- Headache
- Dizziness or Vertigo
- Weakness (generalized, proximal, or distal)
- Syncope and Near Syncope
- Seizures
- Tremors or involuntary movements

Neurological Examination

- Start with taking the patient's history
- Do they have neurologic disease?
- If so, what is the localization of the lesion or lesions?
- Are the findings symmetrical?
- What is the pathophysiology of the abnormal findings?
- What is the preliminary differential diagnosis?

Mental Status Examination

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- Appearance and behavior
- Speech and language
- Mood
- Thoughts and perceptions
- Cognitive function

Glasgow Coma Scale

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Activity		Score
Eye Opening		
None	1 = Even to supraorbital pressure	
To pain	2 = Pain from sternum/limb/supraorbital pressure	
To speech	3 = Nonspecific response, not necessarily to command	
Spontaneous	4 = Eyes open, not necessarily aware	---
Motor Response		
None	1 = To any pain; limbs remain flaccid	
Extension	2 = Shoulder adducted and shoulder and forearm internally rotated	
Flexor response	3 = Withdrawal response or assumption of hemiplegic posture	
Withdrawal	4 = Arm withdraws to pain, shoulder abducts	
Localizes pain	5 = Arm attempts to remove supraorbital/chest pressure	
Obeys commands	6 = Follows simple commands	---
Verbal Response		
None	1 = No verbalization of any type	
Incomprehensible	2 = Moans/groans, no speech	
Inappropriate	3 = Intelligible, no sustained sentences	
Confused	4 = Converses but confused, disoriented	
Oriented	5 = Converses and is oriented	---
		TOTAL (3-15)

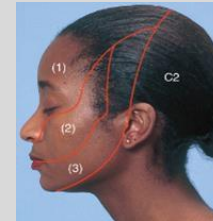
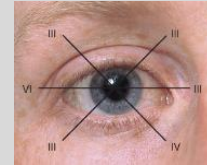
A patient with a score of
3-8 is usually considered to
be in a coma

Summary of Cranial Nerves I-XII

Cranial Nerve	Function of Cranial Nerve	Abnormal Clinical Finding
CN I- Olfactory (Sensory)	Smell	Anosmia
CN II- Optic (Sensory)	Visual acuity, visual field, ocular fundus	Amaurosis
CN III- Oculomotor (Motor)	Ocular movement, lid elevation, pupil constriction, accommodation	Diplopia, ptosis, loss of accommodation
CN IV- Trochlear (Motor)	Extra-ocular movement, downward, internal rotation of eye	Diplopia
CN V- Trigeminal (Motor and Sensory)	Facial, scalp and teeth sensation, corneal reflex (sensory), jaw movement, voice/speech	Weakness of jaw, numbness of face
CN VI- Abducens (Motor)	Extra ocular movement, lateral deviation of the eye	Diplopia
CN VII- Facial (Motor and Sensory)	Taste, general sensation of palate larynx and external ear, lacrimal, submandibular and sublingual gland secretion, facial movements- including facial expression, closing the eye and closing the mouth	Dry mouth, paralysis of facial muscles, loss of lacrimation, loss of taste to anterior 2/3 rd of tongue

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Summary of Cranial Nerves I-XII

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Cranial Nerves	Function of Cranial Nerve	Abnormal clinical finding
CN VIII- Acoustic or Vestibulocochlear (Sensory)	Hearing, balance/equilibrium	Vertigo, tinnitus, deafness, nystagmus
CN IX- Glossopharyngeal (Motor and Sensory)	Taste, sensation of pharynx, and ear, palate elevation, gag reflex (sensory)	Loss of taste in posterior 1/3 rd of tongue, partial dry mouth, pharynx anesthesia
CN X- <u>Vagus</u> (Motor and Sensory)	Taste, sensory to pharynx, larynx, ear, swallowing, phonation; motor to palate, pharynx and larynx; gag reflex (motor); parasympathetic innervation of heart and abdominal viscera	Dysphagia, hoarseness, paralysis of palate
CN XI- Spinal Accessory (Motor)	Head, neck and shoulder movement, phonation	Weakness of head, neck, shoulder, hoarseness
CN XII- Hypoglossal (Motor)	Tongue symmetry and position, voice/speech	Weakness, tongue wasting



Bates 12th Edition Chapter 17
Swartz 8th Edition Chapter 21

CN I- Olfactory (Sensory)

- Ask the patient to close their eyes and occlude one nostril
 - Test substance is brought close to the unobstructed nostril
 - Patient is instructed to sniff and identify the substance
 - Repeat on the other side
- *Test substance must be non-irritating
(ex: fresh ground coffee, lavender, cloves, vanilla beans)



The Examination of the Eye

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- Visual Acuity- CN II (Optic)
- Visual Field by Confrontation- CN II
- Pupil Reaction/Response- CN III (Oculomotor)
- Extra-Ocular Muscle Movement- CN III (Oculomotor), IV (Trochlear), VI (Abducens)
- Anterior and Posterior Segments- Fundusoscopic Examination



Visual Acuity- CN II- Optic (Sensory)

Terminology:

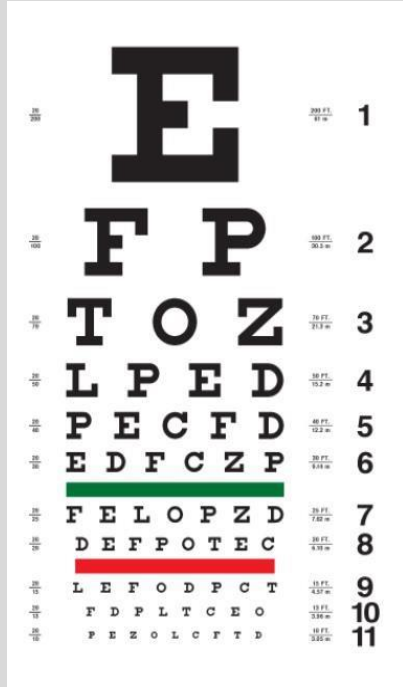
VA-Visual Acuity

OD- Ocular dexter (right eye)

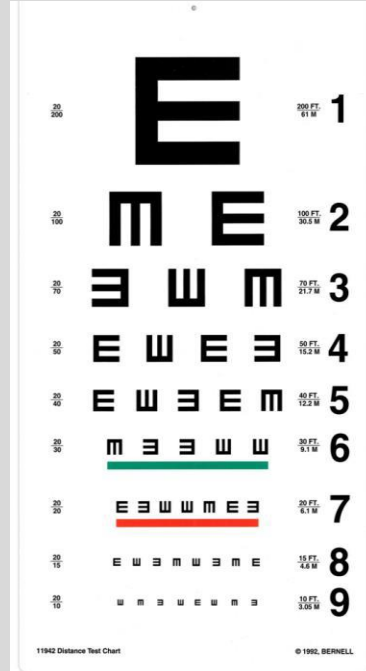
OS- Ocular sinister (left eye)

OU- Oculus uterque (both eyes)

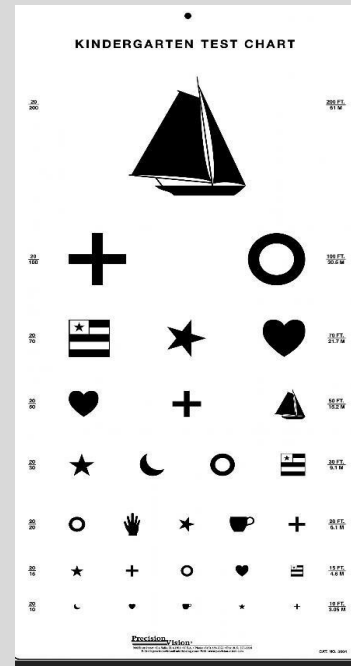
Eye Charts



Snellen



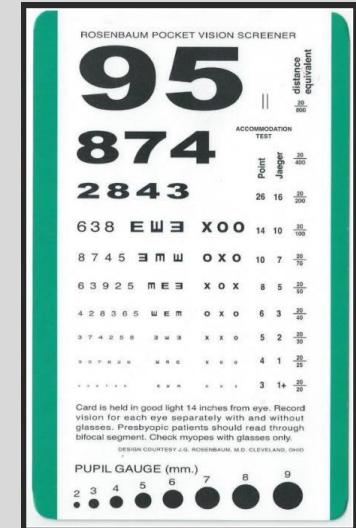
Tumbling E



Kindergarten

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Rosenbaum Pocket Chart

Distance Visual Acuity Testing

- Nomenclature

Distance between the patient and the eye chart

Distance at which the letter can be read by a person with normal visual acuity

Normal: 20/20

Below Normal: 20/40 or higher

Common Refractive Errors

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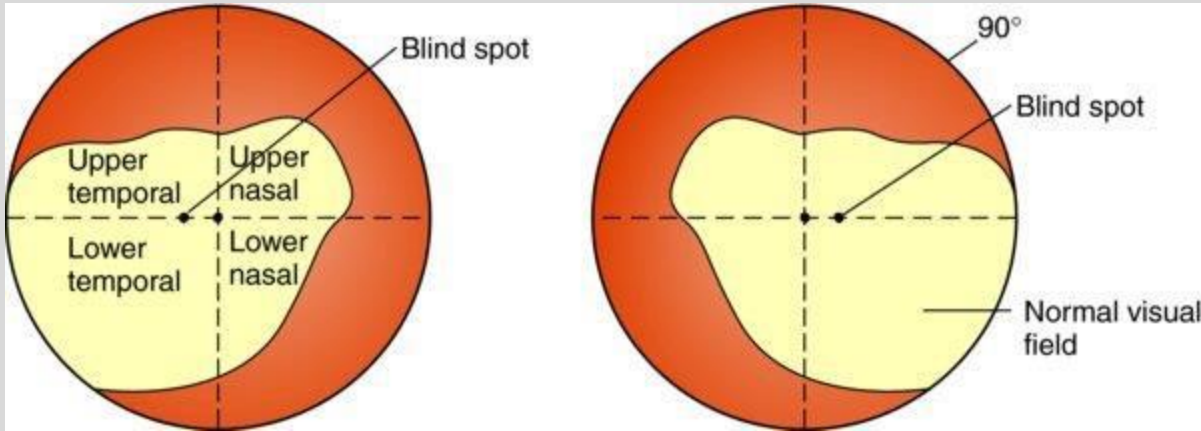
Refractive error	Description
Hyperopia	Globe is too short. Converging lens is used to increase refractive power. Light rays focus behind the retina.
Myopia	Globe is too long. Light rays focus in front of the retina. Distant objects require diverging lens.
Astigmatism	Corneal surface is not perfectly spherical. Refractive power is different at different axes. Needs cylindrical corrective lens.
Presbyopia	Lens becomes unable to increase refractive power to accommodate on near objects.

Source: Ophthalmologic Evaluation. Vaughan & Asbury's General Ophthalmology, 18th edition, Chapter 2

Visual Fields

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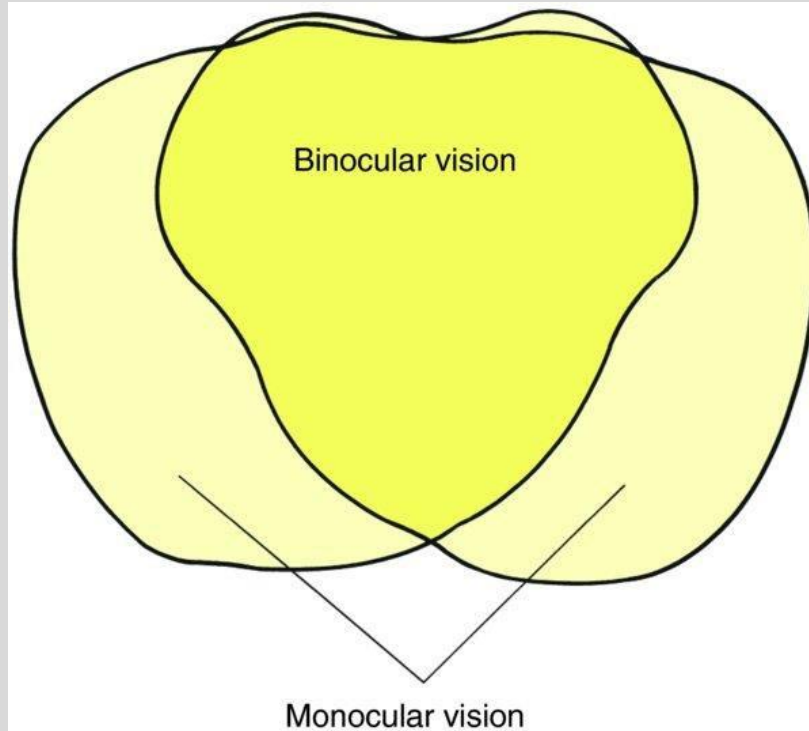


- Entire area seen by an eye when it looks at a central point
- The field extends farthest on the temporal side

Convergence of Visual Fields

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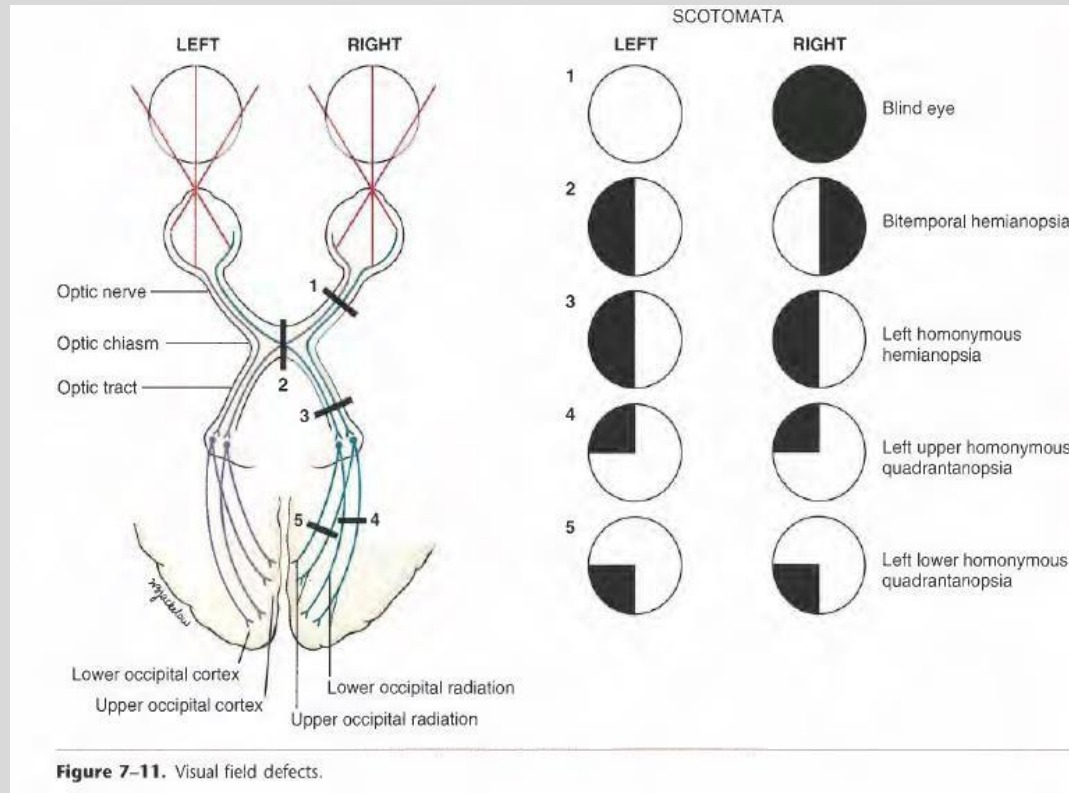
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Visual Field Defects

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Visual Fields by Confrontation CN II

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Static Finger Wiggle Test

- Have the patient look into your eyes.
- Place hands 2 feet apart lateral to patient's ears
- Wiggle both fingers and simultaneously bring them slowly forward as if a bowl. (curving inwards toward central vision)
- Have the patient inform you as soon as movement is seen.
- Repeat for all 4 quadrants of vision.



Visual Fields by Confrontation

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Further testing:

- If a defect is found with static finger wiggle test
- Test one eye at a time by having patient covering their eye with their hand.
- The examiner will cover their opposite eye as well.
- Have patient keep gaze with you and a point equal distant between the examiner and patient move wiggling fingers in the defect area until seen.
- Repeat steps in different levels to “map out” the defect



Inspection

- Stand in front of the patient at eye level
- Assess symmetry, position of the eye and alignment
- Inspect the eyebrow- look for symmetry, sparing of hair
- Inspect the eyelids- ptosis, swelling, lesions, eversion or inversion of the lids



Inspection

Conjunctivitis



Exophthalmos



Ptosis



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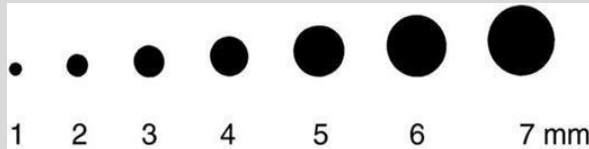
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Assessment of Pupil

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- Assess pupil size and shape
- Pupil sizes may be measured using a card with black circles of varying measurements from 1mm to 7mm.
- If $>5\text{mm}$ - considered large
- If $<3\text{mm}$ – small



Pupil Sizes

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Normal



Dilated



Narcotic Analgesics –
Heroin, Pain Pills



Meth, Cocaine, Ritalin, Diet
Pills, Hallucinogens

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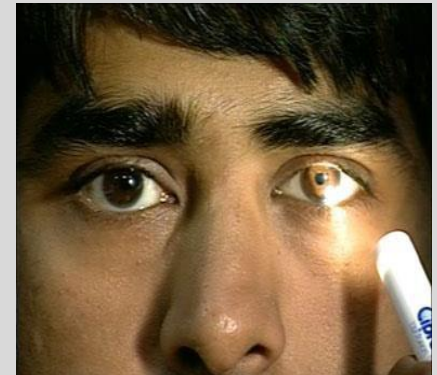
Pupillary Response- CN III- Oculomotor (Motor)

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Direct penlight into eye while patient looking at a distance

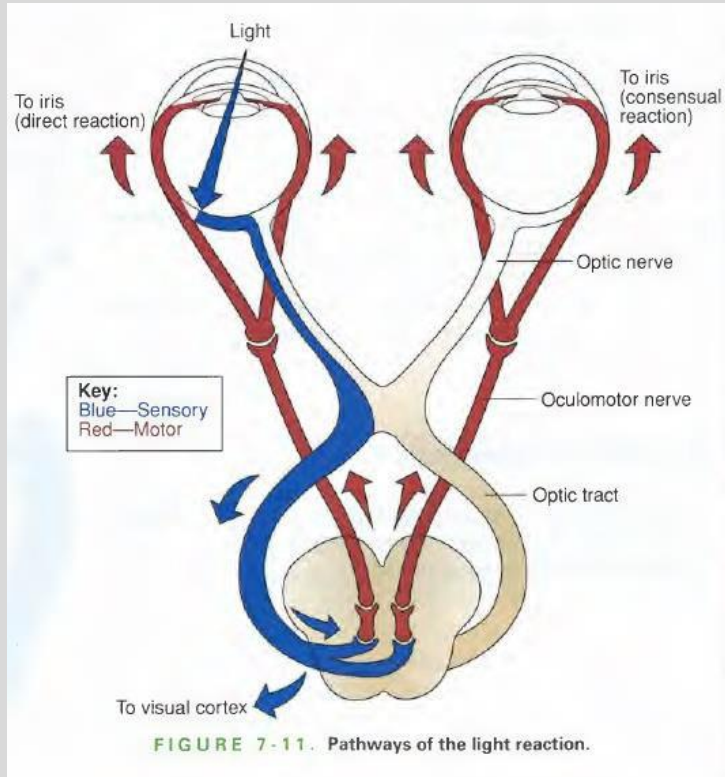
- Confirm that the pupils respond to light
- Compare pupillary diameters
- Direct Response
 - Constriction of ipsilateral eye
- Consensual Response
 - Constriction of contralateral eye



Pupillary Response

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Near Reaction/Convergence

Near reaction: Pupillary constriction with near effort (use if reaction to light is impaired or in question)

- In normal room light, hold a pencil or your finger 10 cm from the patient's eye.
- Ask the patient to alternate looking at an object on a far wall and then at the object in your hand.
- Watch for pupillary constriction with near effort.

Convergence: The ability of the two eyes to work synchronously together

- Ask the patient to follow your finger or object that is approximately 12 inches from the nose in toward 5-8 cm of the nose.
- Normal would be if the converging eyes can follow the object to that point.

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Extraocular Movements CN III, IV, VI

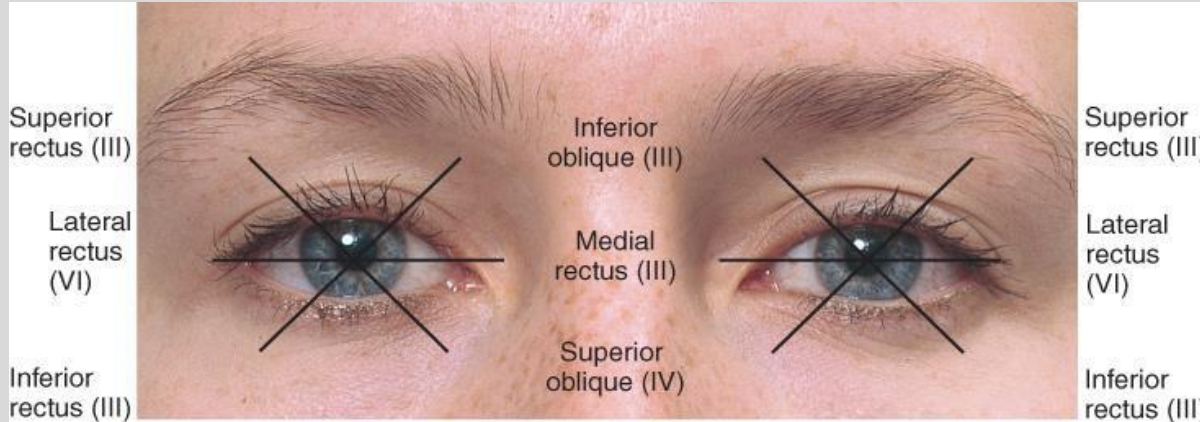
CN III- Oculomotor (Motor)

CN IV- Trochlea(Motor)

CN VI- Abducens (Motor)

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Extraocular Movement Testing

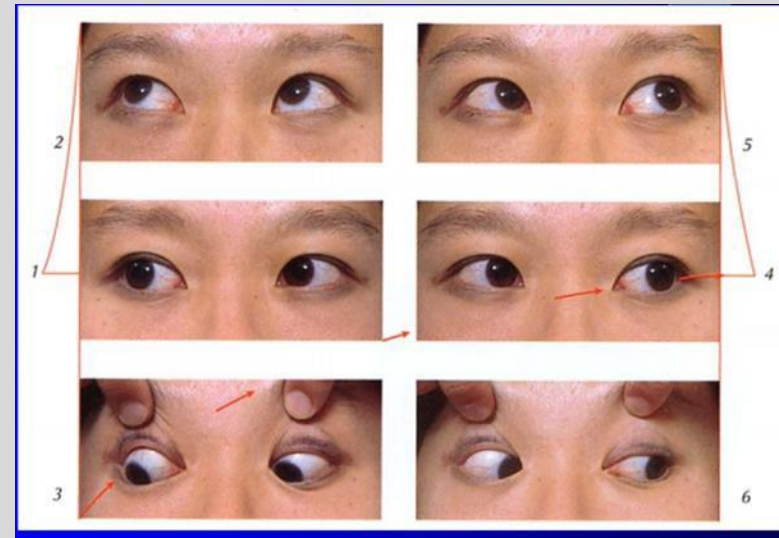
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- Stand in front of the patient
- Ask patient to follow your finger with their eyes only, without turning or moving their head
- Make a wide “H” pattern in the air which covers all 6 cardinal directions of gaze



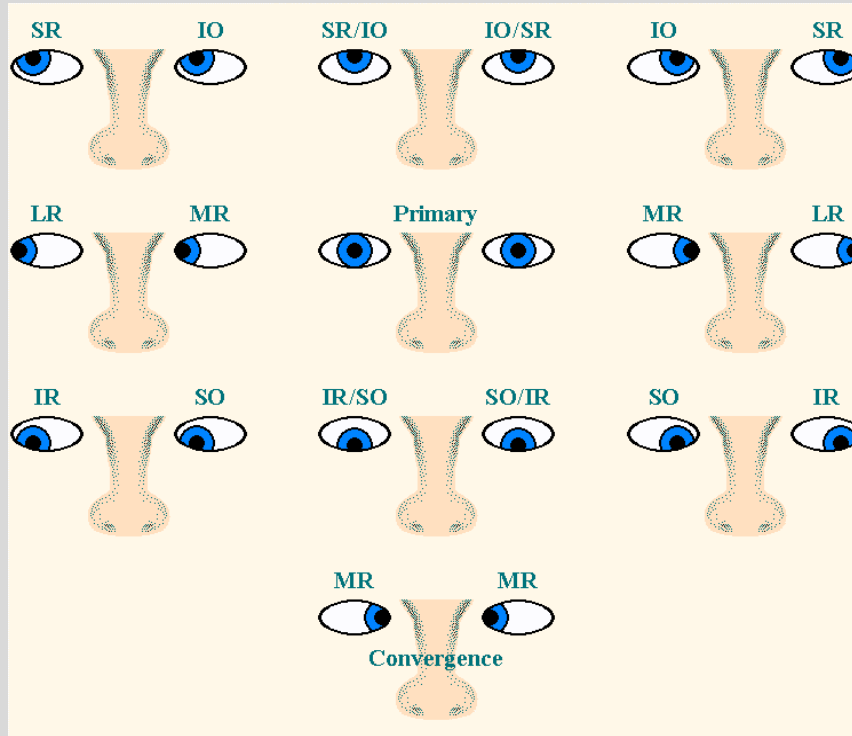
Bates 13th Edition Chapter 12



Ocular Motility

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Source: Ophthalmologic Evaluation. Vaughan & Asbury's General Ophthalmology, 18th Edition, Chapter 2

Diagnosis?

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A



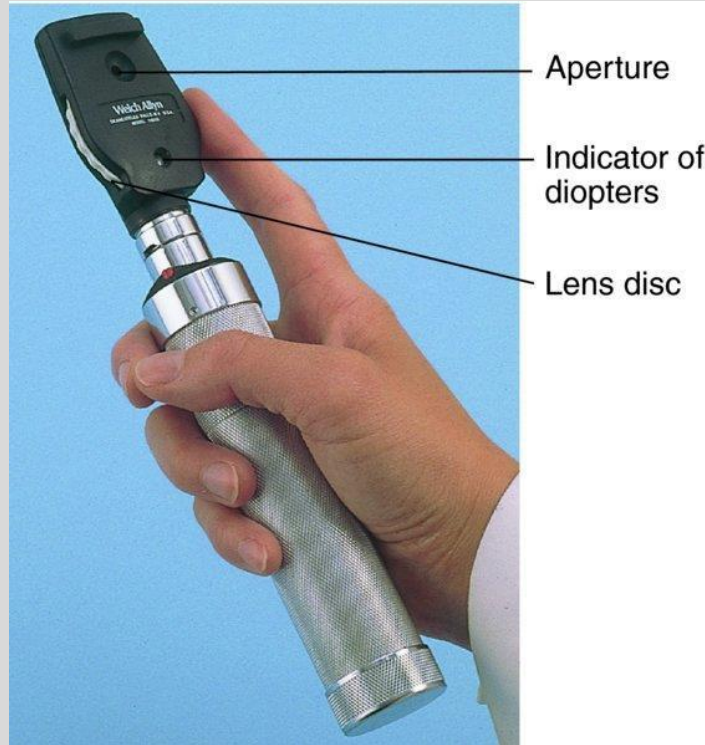
B



C

Fundusoscopic Examination

- Ophthalmoscope



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Fundusoscopic Examination

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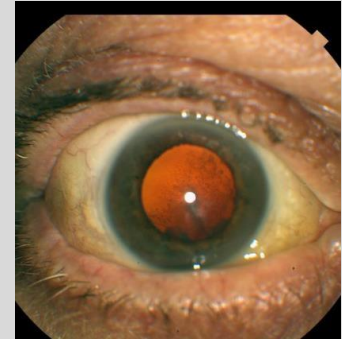
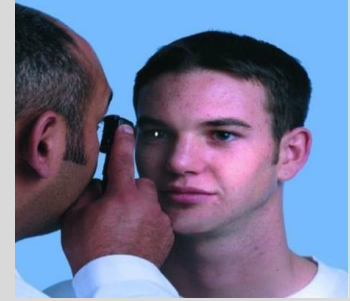
- Examination Room should be darkened
- Lens should be turned onto the large round white beam of light
- Lens Disc is set to 0 Diopter
- Ophthalmoscope is held in the right hand to examine the patient's right eye and in the left hand to examine the patient's left eye.
- Place yourself 15 inches from the patient and 15 degree from the patient's line of vision
- Patient is asked to focus at a distant object.

Fundoscopic Examination

- Place your thumb of your free hand on the patient's eyebrow with the ophthalmoscope against your bony orbit
- When you shine the light into the patient's pupil you will see the orange glow of the pupil- Red Reflex

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Fundoscopic Examination

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- Continue to keep the light beam on the eye and continue the 15 degree angle until you are almost touching the patient's eyelashes.
- Locate the **Optic Disc** by following the blood vessels centrally and noting their enlargement as you approach the disc.
- Focus using the Lens Disc to bring the retina into focus
If the patient is Myopic (nearsighted)- adjust the Lens Disc Counterclockwise (-) red numbers
If the patient is Hyperopic (farsighted)- adjust the Lens Disc Clockwise (+) green numbers

Diopter Power Display →



Fundusoscopic Examination

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- Inspect the **optic disc** for: clarity, color of disc and size of central physiologic cup
- Central physiologic cup is usually yellow-white

Normal



Process

Tiny disc vessels give normal color to the disc.

Papilledema



Process

Elevated intracranial pressure causes intraaxonal edema along the optic nerve, leading to engorgement and swelling of the optic disc.



CN V-Trigeminal- Sensory

- Use a pin or cotton swab
- Assess at the three distributions of the Trigeminal Nerve: Ophthalmic, Maxillary, Mandibular
- Test bilaterally



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Corneal Reflex

- Out of the patient's line of vision, the cornea is touched lightly with a fine wisp of cotton
 - Inspect for closure of bilateral eyelids (blinking) which is a normal response
- Sensory limb of this reflex is with CN V
- Motor response for both sides is with CN VII



CN V- Trigeminal -Motor

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- Palpate the temporal and masseter muscles bilaterally and ask the patient to clench their teeth and note the strength of the contraction
- Have the patient open and move the jaw side to side



CN VII- Facial (Sensory and Motor)

- Sensory- Taste from the Anterior two-thirds of the tongue
- Motor- Inspect the face for asymmetry

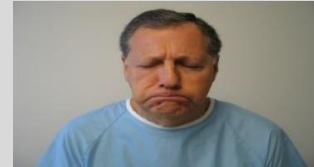
Ask patient :

Raise Eyebrows



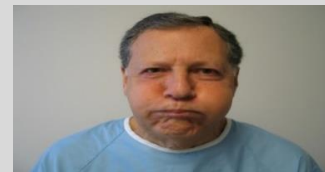
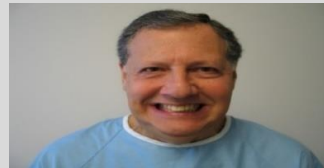
Close both eyes tightly- do
not let me open them

Smile



Frown

Show teeth

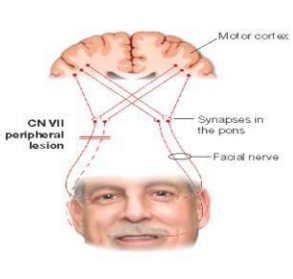
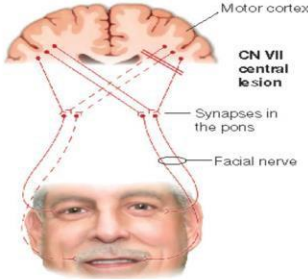


Puff cheeks

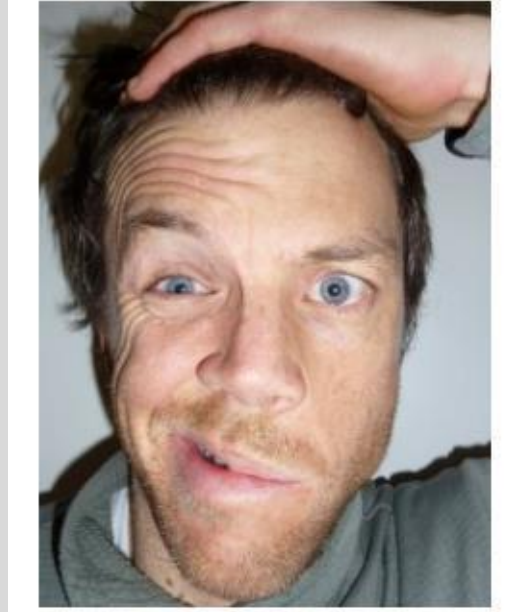
Facial Paralysis Secondary to Peripheral and Central Lesions of CN VII

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CN VII—Peripheral Lesion	CN VII—Central Lesion
Peripheral nerve damage to CN VII paralyzes the entire right side of the face, including the forehead.	Central nerve damage to CN VII paralyzes the lower face but cortical innervation to the forehead is preserved.
	
Closing Eyes Eye does not close; eyeball rolls up Flat nasolabial fold	Closing Eyes Eye closes; perhaps with slight weakness Flat nasolabial fold
Raising Eyebrows Forehead not wrinkled; eyebrow not raised Smiling Paralysis of lower face	Raising Eyebrows Forehead wrinkled; eyebrow raised Smiling Paralysis of lower face

Diagnosis?



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CN VIII- Acoustic or Vestibulocochlear (Sensory)

Gross Auditory Acuity

- Instruct patient to cover one ear
- Examiner whispers word or numbers into the uncovered ear from approximately 2 feet away
- Instruct patient to repeat the word/numbers
- Compare bilaterally

Alternative

- Instruct patient to cover one ear
- Examiner rubs thumb and forefinger together next to the uncovered ear and asks the patient if a sound was heard
- Compare bilaterally

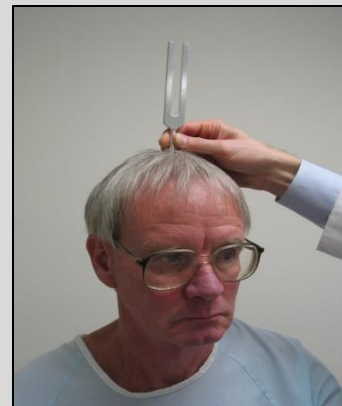


Conductive vs Neurosensory Hearing Loss

Weber Test

Test for lateralization of sound

- Use a tuning fork of 512 Hz
- Place vibrating tuning fork on top of the patient's head
- Ask the patient to identify from which ear that they hear the sound.



Conductive versus Neurosensory Hearing Loss

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Rinne Test- compare air conduction (AC) with bone conduction (BC)

- Use a 512 Hz tuning fork
- Strike the tuning fork to create a vibration
- The base of the vibrating tuning fork is placed against the mastoid process behind the ear that is being evaluated
- Ask the patient to inform you when they no longer hear the sound
- Move the tuning fork lateral to the patient's ear without re-striking it
- Ask the patient if they hear the tuning fork and to inform you when they no longer hear it.



Rinne Test

- Normal- Air Conduction is greater than Bone Conduction
- Unilateral Sensorineural Loss- $AC > BC$
- Conductive Loss- $BC = AC$ or $BC > AC$

Hearing Pathways

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Example- Weber test- sound lateralizes to the left ear

A. Rinne test- BC > AC in the left ear

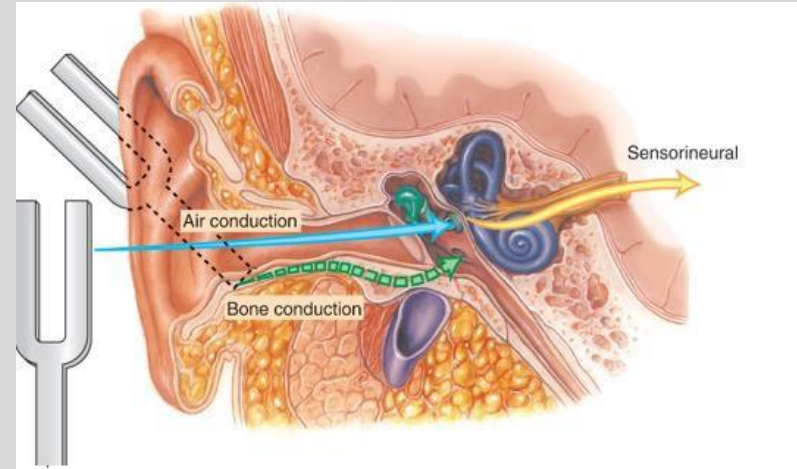
AC > BC in the right ear

Conclusion- There is a conductive loss (cerumen or foreign body) in the **left** ear. (It is thought that there is a decrease in the masking effect of the background noise with conductive hearing loss *)

B. Rinne test- AC > BC in the left ear

AC > BC in the right ear

Conclusion- There is a sensorineural deficit in the **right** ear



Bates 13th Edition Chapter 13

*Swartz 8th Edition Chapter 11- The Ear- Weber Test

<http://arktos.nyit.edu/login?url=https://www.clinicalkey.com/#!/content/book/3-s2.0-B9780323672924000115>

CN IX and X

CN IX-Glossopharyngeal (Sensory & Motor)

Sensory- Gag reflex; Taste for posterior 1/3 of the tongue

CN X- Vagus (Sensory and Motor)

- Listen to the quality of the patient's voice
- Have the patient swallow and observe
- Instruct the patient open their mouth and say "ah"
Observe for: symmetrical movement of the
soft palate and location and movement of the
uvula
- Check for the gag reflex by lightly stimulating the back of the throat using a tongue depressor on each side of the posterior pharynx.

CN XI- Spinal Accessory (Motor)

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- Observe and palpate for atrophy or fasciculations in the trapezius muscles bilaterally
- Ask the patient to shrug their shoulder against your resistance.



AND

- Place your hand on one side of the patient's face and ask the patient to turn their head towards that side.
- Apply resistance



CN XII- Hypoglossal (Motor)

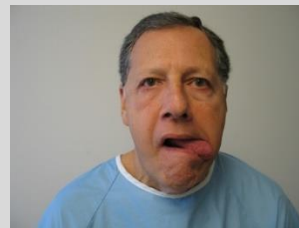
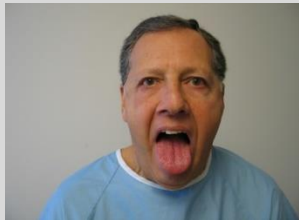
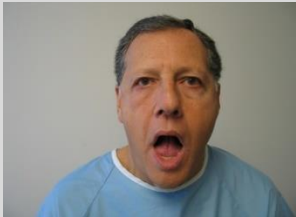
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- Instruct the patient to open their mouth
- Inspect the tongue as it lies on the floor of the mouth
- Instruct patient to protrude the tongue

Note any asymmetry, atrophy or deviations from the midline

- Instruct the patient to move the tongue from side to side
- Have patient push the tongue against the inside of each cheek as the examiner palpates externally for strength



Diagnosis?

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Documentation of a “Normal” Neurological Examination

Neurologic Examination

- Mental Status: The patient is alert, attentive, and oriented to person, place and time. Speech; clear and fluent with normal comprehension

- Cranial Nerves:

CN I- Not tested

CN II- Visual fields are full to confrontation. Fundoscopic exam is normal with sharp discs and no vascular changes. Visual acuity is 20/20 bilaterally

CN III, IV, VI- Pupils 4 mm and reactive to light; extraocular muscles intact (EOMI), no ptosis

Documentation of a “Normal” Neurological Examination

- CN V- Facial sensation is intact to pinprick in all 3 divisions bilaterally, Corneal responses are intact
- Temporal and Masseter strength intact
- CN VII- Face is symmetric with normal eye closure and smile
- CN VIII- Hearing intact bilaterally to rubbing fingers (or whispered voice)
- CN IX, X- Palate elevates symmetrically, phonation is normal, gag reflex intact
- CN XI- Head turning and shoulder shrug intact
- CN XII- Tongue midline, no atrophy

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Documentation in a SOAP note

Vitals- BP- 110/70, Pulse 70, RR 16, Temp 98.6F, Ht 5'4", Wt 140 lb BMI- 24

General appearance- Pt is awake, alert and oriented X 3. appears stated age is well groomed, good hygiene and is in no acute distress

HEENT

- Head: Normocephalic/Atraumatic, No lesions noted
- **Eyes- Visual acuity 20/20 bilaterally. anicteric, conjunctiva pink. Pupils are 4 mm constricting to 2 mm, equally round and reactive to light and accommodation (PERRLA), Extra ocular muscles are intact (EOMI), Disc margins sharp; no hemorrhages or exudates, no arteriolar narrowing**
- Ears- Acuity good to whispered voice, tympanic membranes intact, with good cone of light bilaterally. Weber midline, AC > BC
- Nose- Nasal mucosa pink, no discharge, septum midline and intact; no sinus tenderness
- Throat- Oral mucosa pink, no pharyngeal erythema or exudates, dentition good, Tonsils not enlarged (or absent).
- Neck- supple, full range of motion, no lymphadenopathy, no thyroid enlargement or masses, no abnormal neck (jugular) vein distention

Preparation for DPR Neuro 1 Lab

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- Please be dressed in DPR lab attire
- Bring a charged Ophthalmoscope, Visual Acuity card (Rosenbaum card), Penlight and Tuning Fork
- Review the Neurological DPR Skills Sheet for mental status, the cranial nerves and HEENT Skills Sheet (Eye examination)

References

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- Bates Guide to Physical Examination and History Taking, 11th, 12th Editions, Lynn S. Bickley, MD, FACP, Chapter 7 (Head and Neck), Chapter 17 (Nervous System)
- Bates Guide to Physical Examination and History Taking, 13th Edition, Lynn S. Bickley, MD, FACP, Chapter 9 (Mental Status Exam), Chapter 12 (Eye) and Chapter 13 (Ear and Nose) and Chapter 24 (Nervous System)
- DeJong's the Neurologic Examination, 8th Edition, William W. Campbell MD and Richard J. Barohn, MD, Chapter 20 (The Hypoglossal Nerve)
- Textbook of Physical Diagnosis History and Examination, 8th Edition, Mark H. Swartz, MD, FACP, Chapter 10 (Eye), Chapter 11 (Ear and Nose), Chapter 21 (Nervous System)

Thank you

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